INTERVEHICULAR NETWORKING

Ananthakrishnan R, Arjun S, Rahul S, Viswanth V S, Mrs. Sandhya.P

Department of Electrical and Electronics Engineering, Mar Baselios College of Engineering and Technology, Mar Ivanios Vidyanagar, Nalanchira, Trivandrum, Kerala-695015

ananthurg77@gmail.com, arjunsreekuttan88@gmail.com, rahulsuresh765@gmail.com, viswanthvs2016@gmail.com

***Abstract*--- Due to the increase in population, the automobile demand also increases. Thus, the causes of accidents are accelerating as time passes by. The main cause of such problems are reckless overtaking measures, speeding and poor traffic management. Therefore, providing convenient safety measures to vehicles by incorporating inter-vehicular networking technique decreases this immense ratio. Inter-vehicular networking (IVN) systems has the capability to primitively revamp the areas consisting safety, productivity, and contentment of routine road travel. Focusing this project of Inter-vehicular networking, it comprises of different systems including overtaking possibility detection, accident location tracking, automatic braking system and noiseless horn systems which works in coordination with one another efficiently and effectively such that safety is assured.**

Keywords--- arduino;sensor;IVN

I. INTRODUCTION

Annually, 2.3 million people die due to traffic or road related accidents on an average. Without relentless and progressive tackling measures, the cases of traffic accident are hoped to outgrow by 67% by the year of 2022. About 5-11% of the clashes in road/traffic accidents has been caused due to careless overtaking assumptions. To beat this existing difficulty, numerous attempts has been done, one in all of which is the outcome of car, particularly in automobile engineering to branch towards the protection of car.This number is indubitably softened by setting up local cautionary systems through vehicular networking while also reducing blind spots while overtaking. The system also reduces honking at traffic and provides easy user interface between driver and vehicle.



Figure 1: Accident scenario graph over the years.

Automobile networking is one among the foremost essential enabling technologies needed to utilize a multitudinous of applications associated with motorized vehicles, vehicle gridlocks, drivers, riders and jaywalkers. These implications are quite unoriginal and seems a bit far-fetched to see as a goal to a bunch of experimenters and associations. Intelligent Transportation Systems (ITS) aims to focus in the undertaking of vehicles, pull off vehicle pile ups in traffic, aid drivers with safety which is a main cause of concern and other vital subject matters, together with preconditioning the suitability of applications for travelers which are not any longer restrained to testing grounds and research facilities of companies.

In this project of Inter-vehicular networking it comprises of different systems including overtaking possibility detection, accident location tracking, automatic braking system and noiseless horn systems which works effectively in reducing accidents and helps in better understanding of road artefact.

II. STATISTICS

In keeping with the World Health Organization (WHO), 1.2 million deaths are caused due to road accidents annually internationally; quarter of all these deaths are caused by accidental injuries. Likewise, about 50 million people suffer first degree or second degree injuries related to traffic accidents. If defensive controls don't seems to be working out, then road related deaths are probably going to be turning into a third prominent reason for deaths during the 2020's from ninth place in 1990's. .Henceforth upto reduce these accidents due human miscalculations and poor vehicular interaction we introduce this highly efficient Inter-vehicular Networking system.

III. PROPOSED SYSTEM

A .Block diagram

The block diagram of the projected system has been shown in Fig.2. Our prospective system amasses the required inputs from the assorted sensors which is being fed to the main control portion which are capable of processing these signals.

The ultrasonic sensors, GPS module, GSM module and blue tooth module gives input to the Arduino mega 2560 unit. The micro controller is responsible to process these signals to give relevant data to the LCD panel and to other sensors.
The IR transmitter transfers the signal which is then accepted by the IR receiver only to process it from the micro controller to give an LED or Buzzer feedback. The power supply board provides the input power supply to each and every module from an external source.

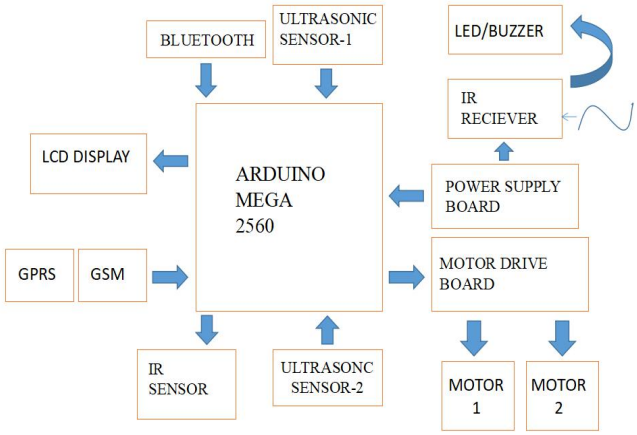


Figure 2: Block diagram of proposed system

The two motors are are responsible for the motion of the prototype and those are connected to the motor driver board.
The system's movement is controlled by a wireless blue tooth module.

B .Components used

i .Arduino micro controller.
The Arduino Mega is a microcontroller board based on the ATMega2560. It has 54 digital input/output pins in which 14 can be used as PWM output, 16 analog inputs, 4 UARTs (Hardware Serial ports), a 16 MHz crystal oscillator, a USB connection, a power jack, an ICSP header and a reset button.

ii .Ultrasonic sensor.
An ultrasonic sensor evaluates the displacement of an object with the aid of ultrasonic sound waves. It has a transducer which can transmit and receive these sound waves that is capable of providing the information about any possible obstacle nearby. High-frequency sound waves reflects from any blokage to supply various echo patterns. The working methodology is similar to that of a radar system.

iii .IR sensor.
The TSOP sensor is a type of IR sensor that has the quality to interpret the output signals like as in television remotes,Home theatre remotes, AC remotes etc. Each and every remote will work within a frequency range of 38kHz and the IC can construe any IR signals and thereby processing them by providing output to pin 3. TSOP-1738 acquires only 38Khz Infrared signals.TSOP-1738 is an Infrared Receiver Sensing element, which can be used to receive Infrared signals of 38Khz. The sensor run on 5V and takes up around 5mA to function

.iv .GPS module
The sensor a satellite-based system which is abbreviated as Global Positioning System, that make use of the satellites and ground stations locate its position on Earth. GPS receivers are broadly employed in smart phones, traffic governing systems, in defence etc. GPS receivers receive packets from 4 satellites for maintaining correct precision and accuracy

v .GSM module.
The GSM modem used is SIM 900 and it mainly consists of 3 connections, RX, TX and the ground. The maximum voltage that can be given to the GSM modem is 12V. It consists of 3 LEDs.ie, the network LED which indicates whether the range is available or not. The board LED indicates whether the board is in proper working condition or not. The ON and OFF LED indicates whether the modem is on or off. Serial connection is usually used. If it is not available, cross connection of RX and TX can be made.

vi .Bluetooth module.
HC-05 is the bluetooth module used here.The bluetooth devices act as the master and the slave. Bluetooth serial compartment allows all sequential enabling devices to intercommunicate with each other using using this..Ranging can be up to <100m which depends upon the sender and recipient, ambiant conditions.

vii .Motor driver board.
The IC used is LM293D . It consists of a 5V ,ground and a 12V pin.It also consists of pins C1A,C1B,C2A&C2B which are used to control the movement of the wheel.

viii .Power supply board.

This is a DC-DC 12V to 3.3volts-5volts-12volts module having a multi-purpose voltage conversion which is perpetually called as the buck converter/step-down voltage converter. The module is capable of neutering the output of the power source before supplying it to the load such that it becomes much safer on delivering the current.

ix . LCD display.

A 16x2 LCD display is used here to demonstrate the readings from the micro controller.

X .Accelerometer.
Accelerometer plays a role in sensing any external forces implying to the vehicle inorder to give the information to the arduino. Arduino then quantifies this piece of information to the threshold value and if it is higher than that, it then arrives into a conclusion that the car met with an accident and triggers the gsm and gps modules to send the accurate position and location to the already linked mobile number. If external force is below rhe threshold value it will not contact the processing unit.

C .Different Systems

i .Accident Location Tracking.

Accident location tracking & Messaging Scheme execution is easy as the system is making use of GSM and GPS technologies. GPS is used with arduino for the purpose of marking the coordinates of the location of the accident while, GSM is used with arduino for the purpose of sending coordinates to the mobile phones. In order to make this system, all controls are activated using the Arduino whereas a LCD is being used for displaying the coordinates. Accident location tracking & Messaging System is very easy and the elements used for this purpose are Vibration Sensor, which recognize the accident and in turn transmits the signals to Arduino. At this period the Arduino gets active and starts to collect the coordinates which are received from the GPS & which are then sent to the Mobile Station using the GSM module. The Vibration Sensor needs 9 Volt of DC supply, if we are choosing a 5 Volt DC power supply, then the circuit will not work correctly. The major units we use here are GSM and GPS modules.

ii .Noiseless Horn.

This transmits the vehicle horn as an IR signals, rather than audible sound. It uses infrared LEDs and TSOPs for transmitting and receiving IR signals as horn. Buzzers used will produce sound in the inner part of the vehicles for making the driver alert. Normal sound horn is also included with silent horn for making the system congruent with the in use system. The driver can use any horn, which depends on the condition, where in most cases, silent horn can be used. This selection is done by 2 switches. If the driver is pressing on the switch gently, the noiseless horn will get activated and if more force is applied by the driver, the sound horn will get activated. We are using two toy cars as our prototype, one for housing the transmitter and the other for housing the receiver. This prototype consists of a transmitter & a receiver module

iii .Automatic Reverse Braking System.

An Ultrasonic setup is placed in back of vehicle and it contains of an emitter and a receiving system. The sensor emitter always emits the ultrasonic waves such that whenever a obstacle is detected while in reverse motion, the waves gets reflected and the receiver receives the signal. The arduino measures the distance by calculating the time taken by the rays to hit the object and reach back such that it is quantified on a mathematical expression : *distance = duration * 0.034/2* . If the distance matches with the preset distance then the brakes are applied automatically along with a message displayed on the LCD panel. If the distance doesn't match with the preset value then the vehicle is good to move backwards without any delay.

iv .Overtaking Assistive System.
For overtaking possibility detection, the ultrasonic sensors used here produces sound waves continuously and checks if, any obstacles are present within a specified range. The reflected sound waves from the obstacle will be received back by the sensor such that by programming it to calculate the distance, the system could easily find the distance between the obstacle and the source. Arduino calculates distance same as in automatic braking system scenario , the only difference is the preset value is changed for longer obstacle detection.

D .Experimental setup

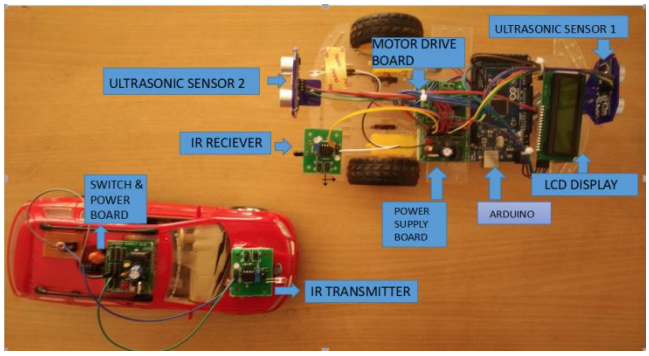


Figure 3:Prototype of the cars

The ultra sonic sensors placed at the front and the back end of the system is responsible for obstacle detection and is displayed in the LCD panel.The sensor at the front detects the opposing vehicle and shows the overtaking chances. While taking the reverse direction ,the sensor at the back detects obstacle and if it is within the specified threshold value, the vehicle stops automatically.TSOP transmitter and receiver is also set upped on the vehicle to execute noiseless horn indication.

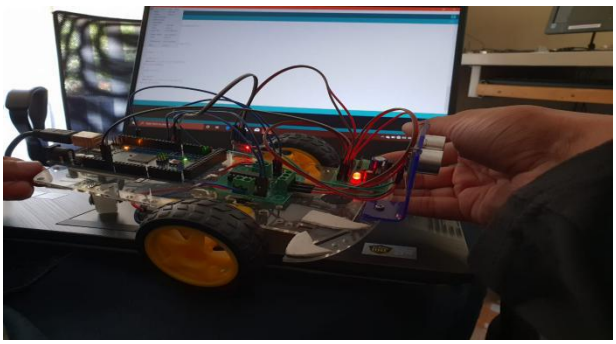
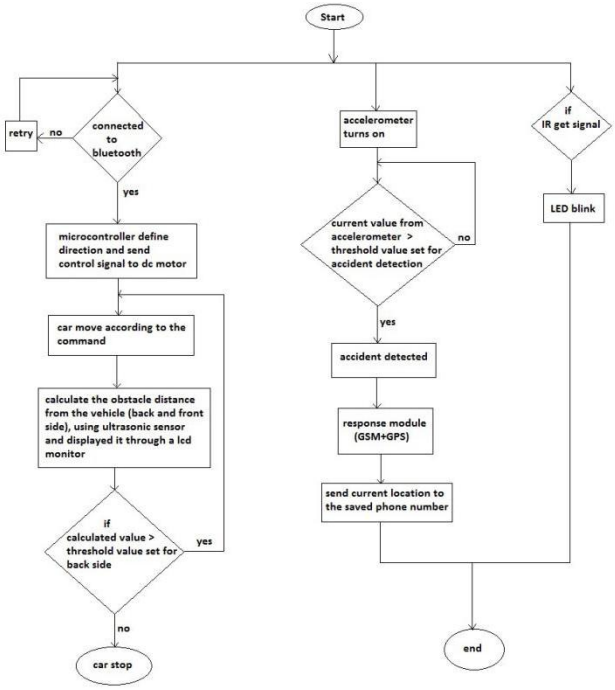


Figure 4: Uploading the coding to the arduino unit and checking for updates.

The vehicle is controlled via Bluetooth and performs forward ,backward ,left and right motions.A suitable power supply is needed which is of 12V is given via an external power source.

IV . FLOWCHART

The above flowchart shows the simplified structure of the complex programming involved with respect to all the sensors and micro controllers.



V. RESULT

The following results were obtained by integrating the different components from testing the system under given conditions.

1. Signals from tsop sensors gives an alert feature to the other vehicle such that noiseless horn working is acquired .
2. Motor driver board and the motor provides movement of the vehicle by controlling it through bluetooth via smartphone.
3.Obstacles are detected while overtaking and close calls are displayed on the LCD panel.
4. The vehicle is automatically stopped while detecting obstacles while in reverse motion avoiding collisions.

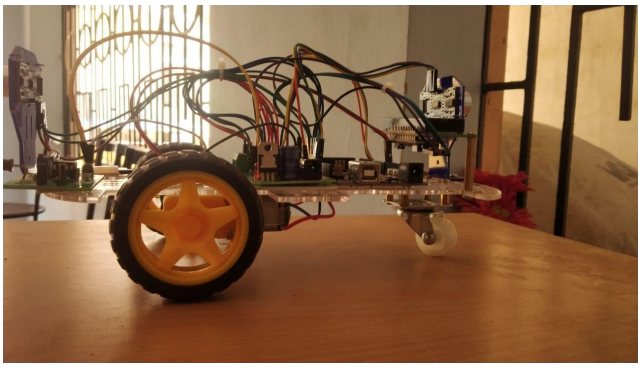


Figure 5: components setup on to the chassis.

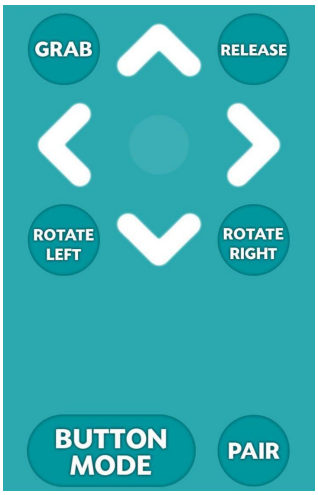


Figure 6: app layout for controlling movements

VI. CONCLUSION

This concludes with summary of automobile networking which we mainly focused on four main systems to minimize the road accidents in our day to day life and to increase a better awareness on road artifacts. By studying and understanding the reasons why most of the accidents occur is one of the reasons why we got motivated to find a solution for this problem and came up with this idea. This project is all about vehicle inbuilt technologies in which more and more systems can be included to overcome all the limitations.
The Inter- vehicular networking improves performance of the network in terms of transmission reliability,

throughput, and interference by introducing cooperation among vehicles. A vehicle-to-vehicle communication code of conduct for collaborative collision warning, overtaking chances and location sensing. Rising wireless technologies for inter-vehicular communications are promising to dramatically decrease the number of fatal roadway accidents by providing early warnings.

VII .REFERENCES

[1] Aakash Choudhary,Arjav Choudhary,Ruban Nersisson. "*GSM based Accelerometer Mounted Accident Detection with Location Tracking and Survivor's Concition Monitoring System",*2019 IEEE International Conference on Distributed Computing,VLSI,Electrical Circuits and Robotics(DISCOVER)

[2] Khalid Anindyaguna, Noor Cholis Basjaruddin, ''*Overtaking Assistant System (Oas) With Fuzzy Logic Method Using Camera Sensor*'', 2016 2nd International Conference Of Industrial, Mechanical, Electrical And Chemical Engineering(Icimece).

[3] Rajiv Kapoor,Rajesh Birok,Divi Sai Manoj. "*Soundless Horn And Remote Patroller",*2014 IEEE International Conference on Vehicular Electronics and Safety.

[4] Fallah Yp, Huang Cl, Sengupta R, Krishnan H. ''*Analysis Of Information Dissemination In Vehicular Ad-Hoc Networks With Application To Cooperative Vehicle Safety Systems*''. IEEETrans Veh Technol. 2011 Jan;60(1):233–247.

[5] J. Luo and J. Hubaux, "*A Survey of Inter-Vehicle Communication*," Technical Report IC/2004/24, EFPL, 2004.

[6] C. Tschudin, H. Lundgren, and E. Nordström, "*Embedding MANETs in the Real World*," Proc. 8th IFIP Int'l. Conf. Personal Wireless Commun., 2003, pp. 578–89.